

Sesi 10

05 Mei 2026

The Influence of Monetary and Fiscal Policy on Aggregate Demand

Pengantar Ekonomi 2

Sumber: Mankiw (Ch 35)

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Monetary and Fiscal Policy

- Monetary policy
 - The open-market operations conducted by the central bank
 - And the interest rate it pays on reserves
- Fiscal policy
 - The levels of government spending and taxation set by the president and Congress

Monetary Policy

Aggregate Demand

- Recall, the **AD** curve slopes downward for three reasons:
 - The wealth effect
 - The interest-rate effect
 - The exchange-rate effect
- Next:
 - A supply-demand model that helps explain the interest-rate effect and how monetary policy affects aggregate demand.

Theory of Liquidity Preference

- The theory of liquidity preference
 - Keynes's theory that the interest rate (r) adjusts to bring money supply and money demand into balance
 - Assumption: **expected rate of inflation is constant**
- The nominal interest rate: is the interest rate as usually reported.
- The real interest rate : is the interest rate corrected for the effects of inflation.
- When there is no inflation
 - The nominal interest rate = the real interest rate

Theory of Liquidity Preference

Money supply, ***MS***

- Assumed fixed by the Fed, does not depend on interest rate

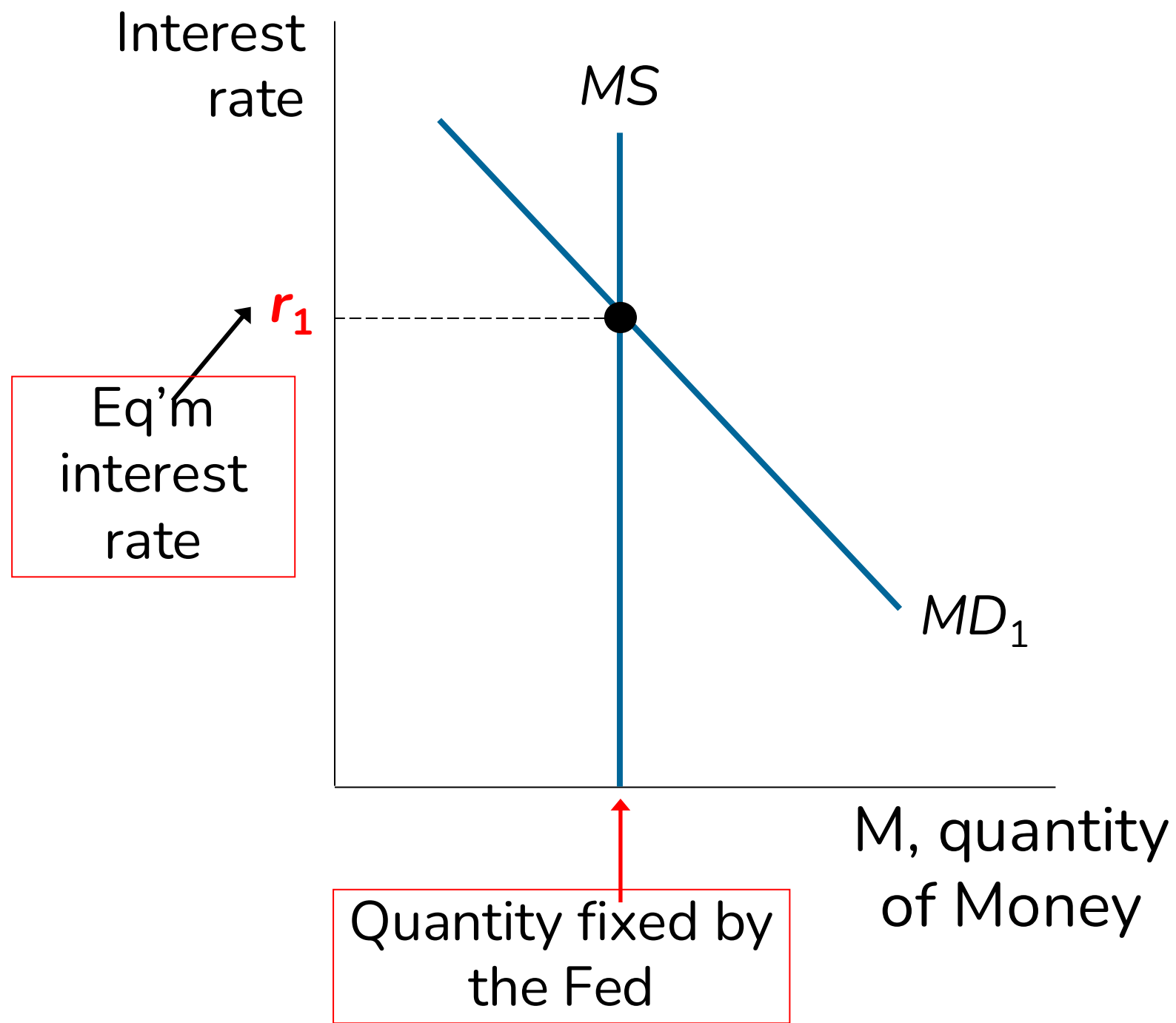
Tool the Fed uses to can change the ***MS***:

- Open market operations
- Change the interest rate it pays on reserves
- Change reserve requirements
- Change the discount rate

Money demand, ***MD***

- Reflects how much wealth people want to hold in liquid form
- Assume household wealth includes only two assets:
 - Money – liquid but pays no interest
 - Bonds – pay interest but not as liquid
- A household's "money demand" reflects its preference for liquidity

How r is determined



MS curve is vertical: Changes in r do not affect MS , which is fixed by the Fed.

MD curve is downward sloping:
A fall in r increases the quantity of money demanded.

- Variables that influence money demand: Y , r , and P .
- Suppose real income (Y) rises:
 - Households want to buy more goods and services, so they need more money
 - To get this money, they attempt to sell some of their bonds.

An increase in Y causes an increase in money demand, other things equal.

Active Learning 1: Determinants of money demand

What happens to money demand in the following two scenarios?

A. Suppose r rises, but Y and P are unchanged.

B. Suppose P rises, but Y and r are unchanged.

Answers:

A. Suppose r rises, but Y and P are unchanged.

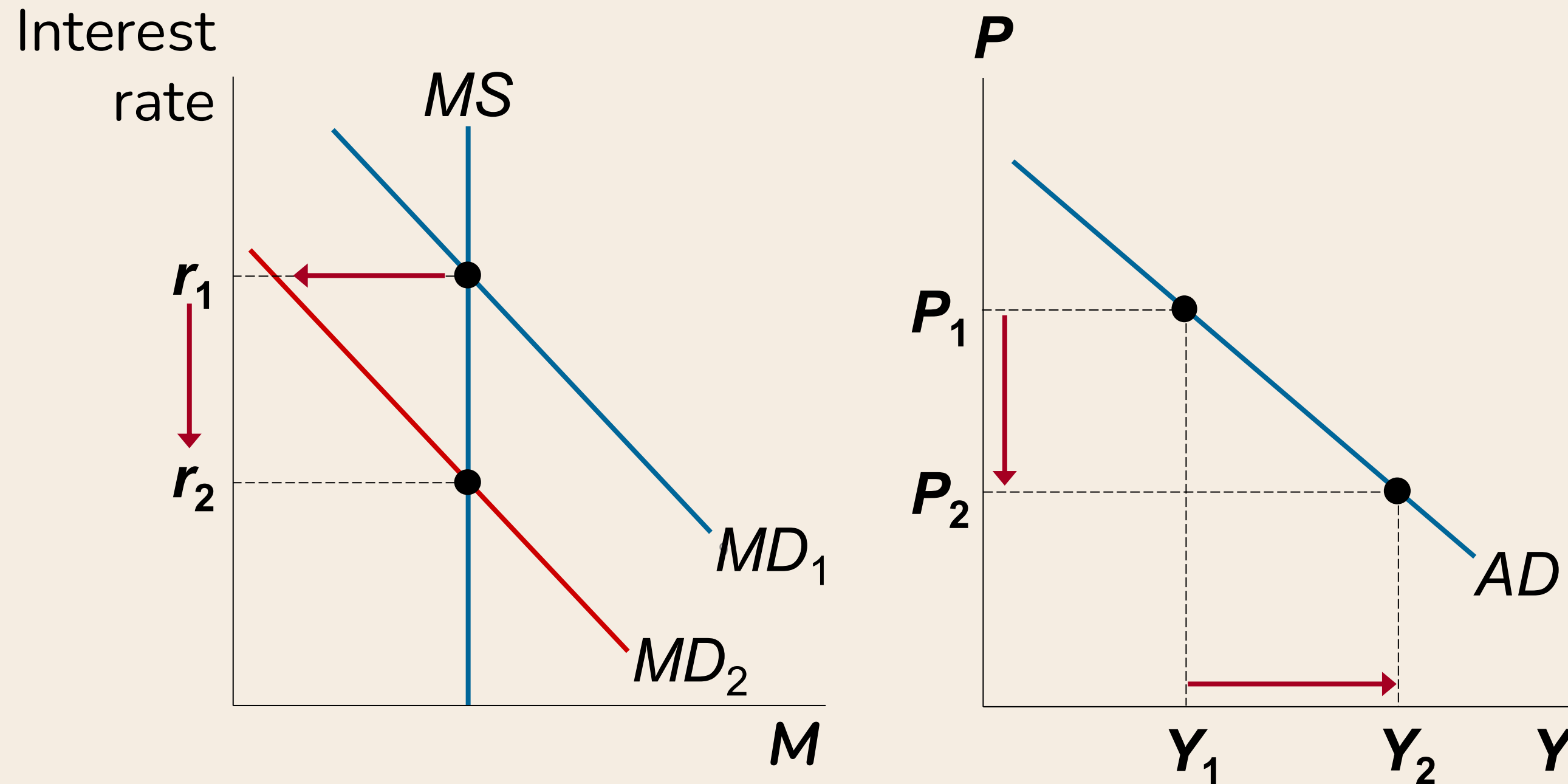
- r is the opportunity cost of holding money.
- An increase in r reduces the quantity of money demanded: households attempt to buy bonds to take advantage of the higher interest rate.
- Hence, an increase in r causes a decrease in the quantity of money demanded, other things equal.

B. Suppose P rises, but Y and r are unchanged.

- If Y is unchanged, people will want to buy the same amount of goods and services.
- Since P is higher, they will need more money to do so.
- Hence, an increase in P causes an increase in money demand, other things equal.

How the interest-rate effect works

A fall in P reduces money demand, which lowers r .



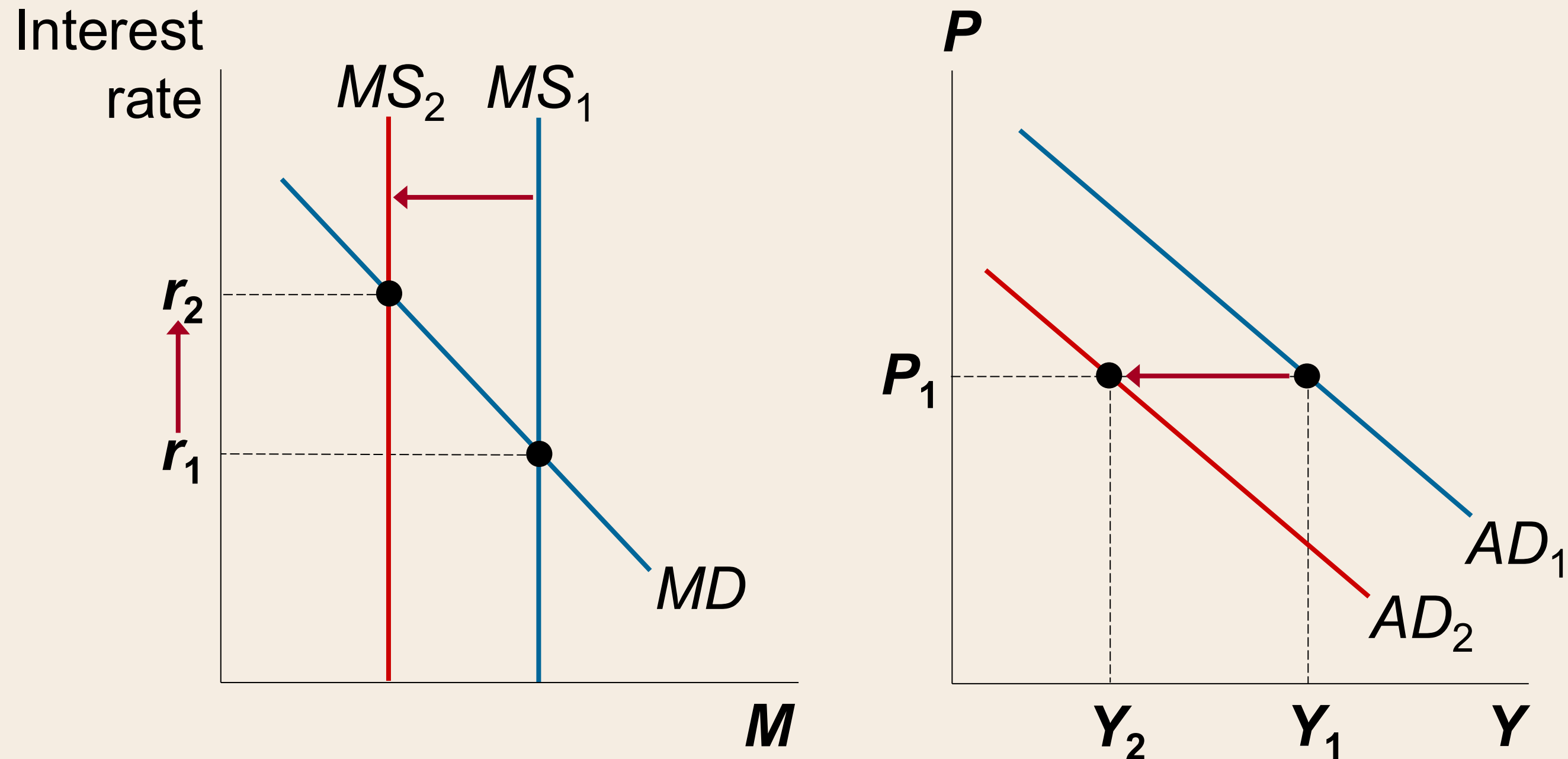
A fall in r increases I and the quantity of g&s demanded, Y .

Monetary Policy and the *AD*

- Shifting the aggregate-demand curve using monetary policy
 - If the Fed decreases the money supply by selling government bonds in open-market operations
 - Interest rate increases to balance money supply and money demand
 - Households and firms respond to the higher interest rate: reduce spending and investment

EXAMPLE 1: Effects of reducing MS

The Fed can raise r by reducing the money supply.



An increase in r reduces the quantity of g&s demanded.

The Role of Interest-Rate Targets

- Because
 - The ***MS*** is hard to measure with sufficient precision and ***MD*** fluctuates over time
 - Leads to fluctuations in interest rates, ***AD***, output
- Fed policy: set a target for federal funds rate
 - Accommodates the day-to-day shifts in ***MD*** by adjusting the ***MS*** accordingly

Monetary policy can be described either in terms of the money supply or in terms of the interest rate.

Active Learning 2: Monetary policy

For each of the events below,

- Determine the short-run effects on output
 - Determine how the Fed should adjust the money supply and interest rates to stabilize output
- A. Congress tries to balance the budget by cutting government spending.
- B. A stock market boom increases household wealth.
- C. War breaks out in the Middle East, causing oil prices to soar.

Answers:

- A. This event would reduce aggregate demand and output.
To stabilize output, the Fed should increase **MS** and reduce **r** to increase aggregate demand.
- B. This event would increase aggregate demand, raising output above its natural rate.
To stabilize output, the Fed should reduce **MS** and increase **r** to reduce aggregate demand.
- C. This event would reduce aggregate supply, causing output to fall.
To stabilize output, the Fed should increase **MS** and reduce **r** to increase aggregate demand.

The Zero Lower Bound

- Liquidity trap
 - If interest rates have already fallen to around zero
 - Monetary policy may no longer be effective, since nominal interest rates cannot be reduced further
 - Aggregate demand, production, and employment may be "trapped" at low levels
- A central bank continues to have tools to expand the economy:
 - Forward guidance: raise inflation expectations by committing to keep interest rates low
 - Quantitative easing: buy a larger variety of financial instruments (mortgages, corporate debt, and longer-term government bonds)
 - Both tools were used by the Fed during the Great Recession and the coronavirus recession)

Fiscal Policy

How Fiscal Policy Influences ***AD***

- Fiscal policy:
 - Setting the level of government purchases (***G***) and taxation (***T***) by government policymakers
 - An increase in ***G*** and/or decrease in ***T***, shifts ***AD*** right
 - A decrease in ***G*** and/or increase in ***T***, shifts ***AD*** left

EXAMPLE 2: The multiplier effect

Assume the government buys \$2 billion of military trucks from Oshkosh Corporation.

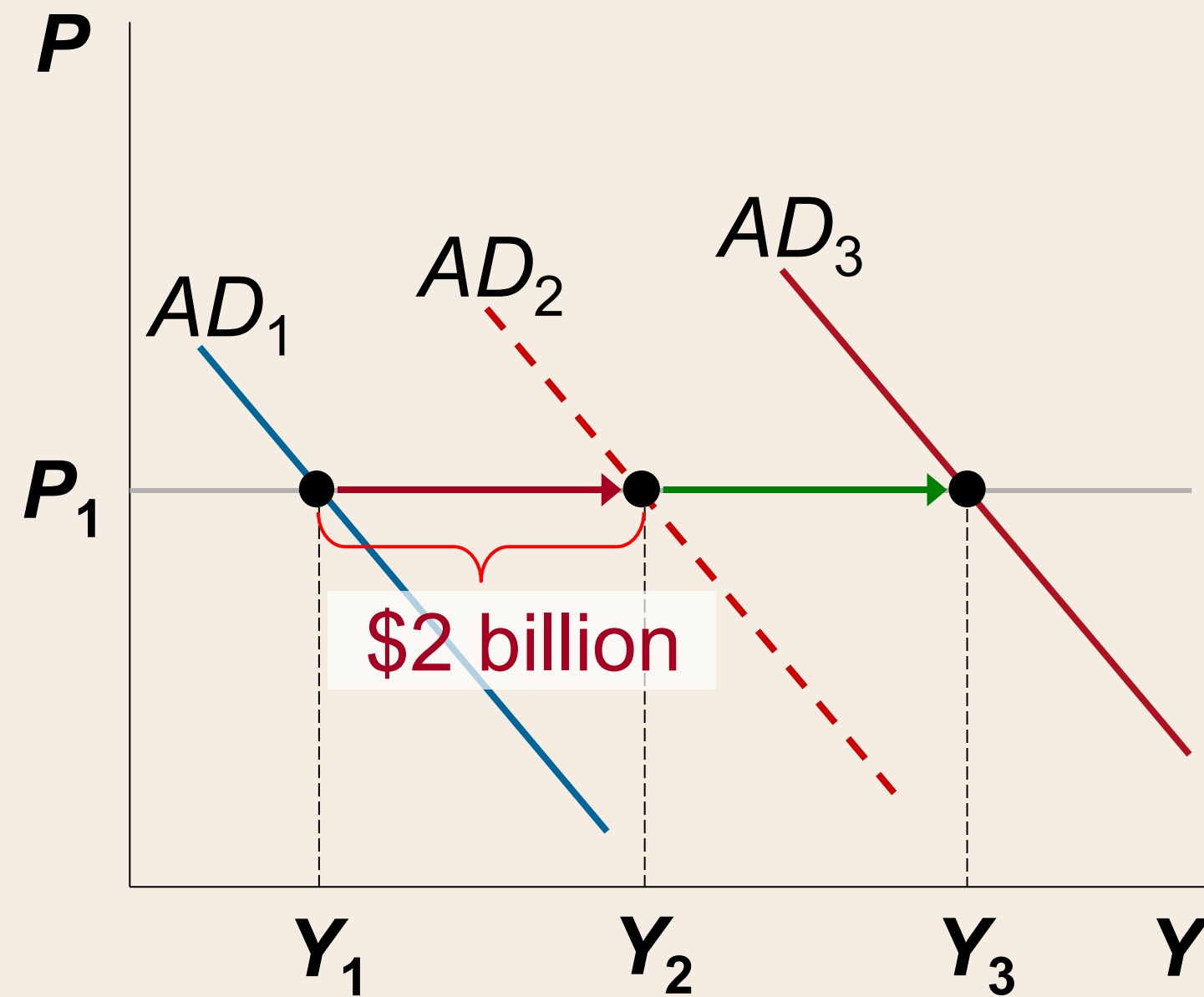
- What is the effect of this purchase on the aggregate demand?
- Because **G** increases by \$2 billion, the **AD** curve shifts to the right by \$2 billion
- Oshkosh's revenue increases by \$2 billion
 - Distributed to Oshkosh's workers (as wages) and owners (as profits or stock dividends).
 - These people are also consumers and will spend a portion of the extra income: **C** increases so **AD** increases

EXAMPLE 2: The multiplier effect diagram

A \$2b increase in **G** initially shifts **AD** to the right by \$2b.

The increase in **Y** causes **C** to rise, which shifts **AD** further to the right.

*Multiplier effect: the additional shifts in **AD** that result when expansionary fiscal policy increases income and thereby increases consumer spending.*



Marginal Propensity to Consume

- How big is the multiplier effect?
 - Depends on how much consumers respond to increases in income.
- Marginal propensity to consume, $MPC = \Delta C / \Delta Y$
 - Fraction of extra income that households consume rather than save
- Example
 - If $MPC = 0.8$ and income rises \$100, C rises \$80.

A formula for the spending multiplier

Notation: $\Delta \mathbf{G}$ is the change in $\mathbf{G}$,
 $\Delta \mathbf{Y}$ and $\Delta \mathbf{C}$ are the ultimate changes in $\mathbf{Y}$ and $\mathbf{C}$

$$\mathbf{Y} = \mathbf{C} + \mathbf{I} + \mathbf{G} + \mathbf{NX}$$

identity

$$\Delta \mathbf{Y} = \Delta \mathbf{C} + \Delta \mathbf{G}$$

$\mathbf{I}$ and $\mathbf{NX}$ do not change

$$\Delta \mathbf{Y} = \text{MPC} \Delta \mathbf{Y} + \Delta \mathbf{G}$$

because $\Delta \mathbf{C} = \text{MPC} \Delta \mathbf{Y}$

solved for $\Delta \mathbf{Y}$

$$\Delta \mathbf{Y} = \frac{1}{1 - \text{MPC}} \Delta \mathbf{G}$$

The multiplier

A formula for the tax multiplier

$$\Delta \mathbf{Y} = - \frac{\text{MPC}}{1 - \text{MPC}} \Delta \mathbf{T}$$

The multiplier

EXAMPLE 3: Calculating the spending multiplier

- Calculate the spending multiplier when **MPC** is 0.5, 0.75, and 0.9.
- What is the relationship between the **MPC** and the simple multiplier?
- If $MPC = 0.5$, multiplier = $1 / (1 - MPC) = 2$
- If $MPC = 0.75$, multiplier = 4
- If $MPC = 0.9$, multiplier = 10
- *The bigger the **MPC**, the bigger the multiplier.*
- A bigger **MPC** means changes in **Y** cause bigger changes in **C**, which in turn cause bigger changes in **Y**.

Other Applications of the Multiplier Effect

- The multiplier effect:
 - Each \$1 increase in **G** can generate more than a \$1 increase in aggregate demand.
 - Also true for the other components of **GDP** (**C**, **I**, **G**, **NX**)

EXAMPLE 4: How big of a change in AD ?

Suppose a recession overseas reduces the demand for U.S. net exports by \$8 billion.

- What is the initial change in AD ?
- If $MPC = 0.8$, what is the change in output?
- Initially, AD falls by \$8 billion
- The spending multiplier = $1 / (1 - MPC) = 5$
- The decrease in Y is = multiplier * change in $NX = 5 * \$8 \text{ billion} = \40 billion

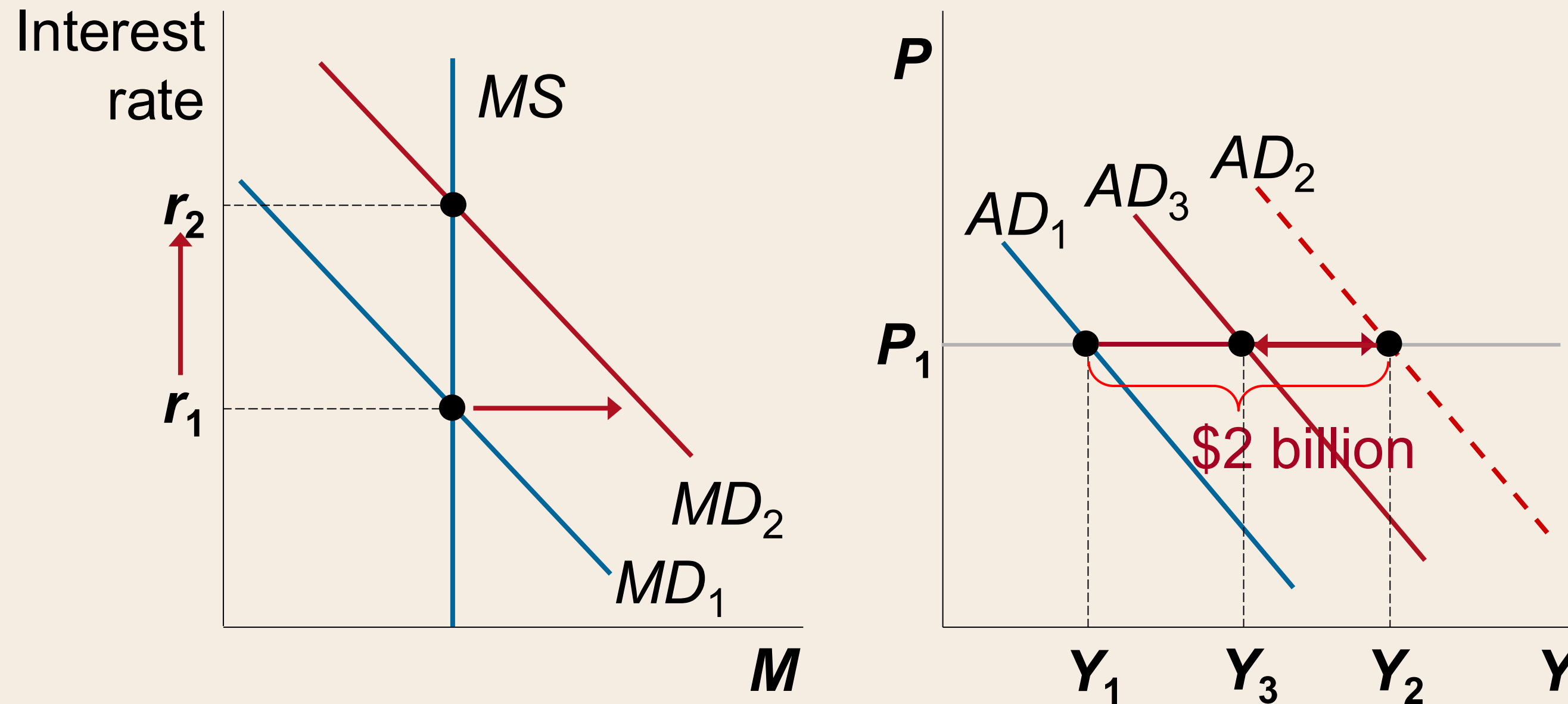
The Crowding-Out Effect

- The crowding-out effect
 - Offset in aggregate demand
 - Results when expansionary fiscal policy raises the interest rate
 - Thereby reduces investment spending
 - Which reduces the net increase in aggregate demand.
 - So, the size of the **AD** shift may be smaller than the initial fiscal expansion.

EXAMPLE 5: The crowding-out effect

Assume the government buys \$2 billion of military trucks from Oshkosh Corporation.

A \$2b increase in G initially shifts AD right by \$2b.



But higher Y increases MD and r , which reduces AD .

Changes in Taxes

- A tax cut
 - Increases households' take-home pay
 - Households respond by spending a portion of this extra income, shifting **AD** to the right
 - The size of the shift is affected by the multiplier and crowding-out effects
- Another factor: households' perception
 - Permanent tax cut – large impact on **AD**
 - Temporary tax cut – small impact on **AD**

Active Learning 3: Fiscal policy effects

The economy is in recession. Policymakers think that shifting the **AD** curve rightward by \$200 billion would end the recession.

- A. If **MPC** = 0.8 and there is no crowding out, how much should Congress increase **G** to end the recession?
- B. If there is crowding out, will Congress need to increase **G** more or less than this amount?

Active Learning 3: Answers

Shifting the **AD** curve rightward by \$200b would end the recession.

- A. If **MPC** = 0.8 and there is no crowding out, how much should Congress increase **G** to end the recession?
- Multiplier = $1 / (1 - \text{MPS}) = 1 / (1 - .8) = 5$
 - Increase **G** by \$40b to shift aggregate demand by:
 $\Delta Y = \text{multiplier} * \Delta G = 5 \times \$40\text{b} = \$200\text{b}.$

Shifting the **AD** curve rightward by \$200b would end the recession.

- B. If there is crowding out, will Congress need to increase **G** more or less than this amount?
- Crowding out reduces the impact of **G** on **AD**.
 - To offset this, Congress should increase **G** by a larger amount.

Fiscal Policy and Aggregate Supply

Fiscal policy can affect aggregate supply

People respond to incentives

- A cut in the tax rate
 - Gives workers incentive to work more, so it might increase the quantity of goods and services supplied and shift AS to the right.
 - People who believe this effect is large are called “Supply-siders”
- **G** might affect aggregate supply.
 - Example: government increases spending on roads.
 - Better roads may increase business productivity, which increases the quantity of goods and services supplied, shifts AS to the right
 - This effect is probably more relevant in the long run: it takes time to build the new roads and put them into use

The Case for Active Stabilization Policy

- Keynes: “Animal spirits” cause waves of pessimism and optimism among households and firms, leading to shifts in aggregate demand and fluctuations in output and employment.
- Also, other factors cause fluctuations,
 - Booms and recessions abroad
 - Stock market booms and crashes
- If policymakers do nothing
 - These fluctuations are destabilizing to businesses, workers, consumers.
- Proponents of active stabilization policy
 - Government should use policy to reduce these fluctuations:
 - When **GDP** falls below its natural rate, use expansionary monetary or fiscal policy to prevent or reduce a recession.
 - When **GDP** rises above its natural rate, use contractionary policy to prevent or reduce an inflationary boom

Case Against Active Stabilization Policy

- Monetary policy affects economy with a long lag:
 - Firms make investment plans in advance, so I takes time to respond to changes in r
 - Most economists believe it takes at least 6 months for monetary policy to affect output and employment
- Fiscal policy also works with a long lag:
 - Changes in G and T require acts of Congress.
 - Legislative process can take months or years
- Due to these long lags
 - Critics of active policy argue that such policies may destabilize the economy rather than help it:
 - By the time the policies affect aggregate demand, the economy's condition may have changed.
 - Contend that policymakers should focus on long-run goals like economic growth and low inflation.

Automatic Stabilizers

- Automatic stabilizers:
 - Changes in fiscal policy that stimulate aggregate demand when economy goes into recession
 - Occur without policymakers having to take any deliberate action

Example:

- The tax system
 - In recession, taxes fall automatically, which stimulates aggregate demand
- Some government spending
 - In recession, more people apply for public assistance (welfare, unemployment insurance)
 - Government spending on these programs automatically rises, which stimulates aggregate demand

THINK-PAIR-SHARE

The news reports that the Fed raised interest rates by a quarter of a percent today to head off future inflation. The report then moves to interviews with prominent politicians. The response of a member of Congress to the Fed's move is negative. She says, "The Consumer Price Index has not increased, yet the Fed is restricting growth in the economy, supposedly to fight inflation. My constituents will want to know why they are going to have to pay more when they get a loan, and I don't have a good answer. I think this is an outrage and I think Congress should have hearings on the Fed's policymaking powers."

- A. What interest rate did the Fed raise?
- B. State the Fed's policy in terms of the money supply.
- C. Why might the Fed raise interest rates before the CPI starts to rise?
- D. Many economists believe that the Fed needs to be independent of politics. Use the congresswoman's statement to explain why so many economists argue for Fed independence

CHAPTER IN A NUTSHELL

- In his work on short-run economic fluctuations, Keynes proposed **the theory of liquidity preference** to explain the determinants of the interest rate. According to this theory, the interest rate adjusts to balance the supply and demand for money.
- **An increase in the price level** raises money demand and increases the interest rate that brings the money market into equilibrium. A higher interest rate reduces investment spending and, thereby, reduces the quantity of goods and services demanded.
- The **downward-sloping aggregate-demand** curve expresses this negative relationship between the price level and the quantity demanded.
- Monetary policy: **increasing money supply** reduces the equilibrium interest rate for any given price level; stimulates investment spending; the aggregate-demand curve shifts to the right.
- Fiscal policy: **increasing government purchases** or a cut in taxes shifts the aggregate-demand curve to the right.
- When the government alters spending or taxes, the resulting shift in aggregate demand can be larger or smaller than the fiscal change.
- **The multiplier effect** amplifies the effects of fiscal policy.
- **The crowding-out effect** dampens the effects of fiscal policy.
- **Advocates of active stabilization policy**: changes in attitudes by households and firms shift aggregate demand; if the government does not respond, the result is undesirable and unnecessary fluctuations in output and employment.
- **Critics of active stabilization policy**: due to long lags, attempts at stabilizing the economy often end up being destabilizing.

Latihan Soal

Ketika pengeluaran pemerintah meningkat maka

- a. permintaan agregat naik sebesar peningkatan pengeluaran pemerintah
- b. permintaan agregat naik sebesar peningkatan pengeluaran pemerintah, namun diikuti penurunan permintaan agregat akibat adanya *crowding-out effect*
- c. permintaan agregat naik dengan besaran yang lebih kecil daripada peningkatan pengeluaran pemerintah
- d. permintaan agregat naik dengan jumlah yang lebih besar daripada peningkatan pengeluaran pemerintah, namun diikuti penurunan permintaan agregat akibat adanya *crowding-out effect*

Misalkan diketahui *marginal propensity to consume* (MPC) adalah 0,75, maka peningkatan belanja pemerintah sebesar Rp120 miliar akan meningkatkan permintaan agregat sebesar

- a. Rp480 milyar, dan efeknya akan lebih besar jika terdapat *crowding-out effect*
- b. Rp480 miliar, dan efeknya akan lebih kecil jika terdapat *crowding-out effect*
- c. Rp160 miliar, dan efeknya akan lebih besar jika terdapat *crowding-out effect*
- d. Rp160 miliar, dan efeknya akan lebih kecil jika terdapat *crowding-out effect*

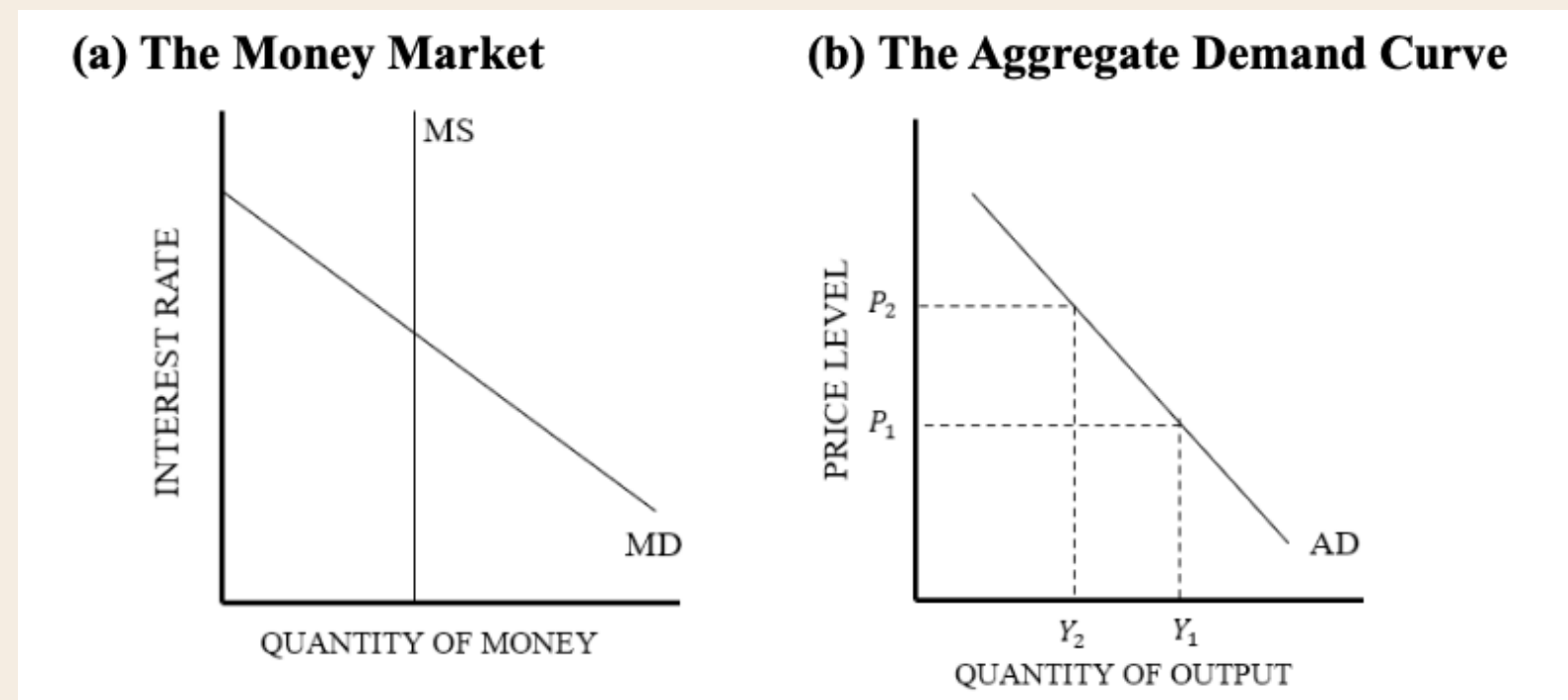
Model preferensi likuiditas (*the liquidity-preference model*) menyatakan bahwa ketika Bank Sentral menurunkan jumlah uang beredar, maka

- a. keseimbangan tingkat bunga akan meningkat
- b. kurva permintaan agregat bergeser ke kanan
- c. kurva penawaran agregat jangka pendek bergeser ke kiri
- d. permintaan barang dan jasa yang diminta tidak berubah pada tingkat harga tertentu

Tujuan dari kebijakan moneter dan kebijakan fiskal adalah untuk

- a. mengimbangi perubahan permintaan agregat sehingga inflasi bisa dihilangkan
- b. mengimbangi perubahan permintaan agregat demi stabilisasi ekonomi
- c. memperkuat perubahan permintaan agregat sehingga tercipta fluktuasi output dan kesempatan kerja
- d. memperkuat perubahan permintaan agregat sehingga pertumbuhan ekonomi meningkat

Gunakan diagram di bawah ini untuk menjawab pertanyaan berikut.



Penurunan permintaan output agregat pada kurva AD dari Y_1 ke Y_2 dapat dijelaskan sebagai berikut.

- Kenaikan tingkat harga menyebabkan kurva permintaan uang bergeser ke kanan; pergeseran MD ini menyebabkan r naik sehingga Y turun.
- Penurunan tingkat harga menyebabkan kurva permintaan uang bergeser ke kanan; pergeseran MD ini menyebabkan r naik sehingga Y turun.
- Kenaikan tingkat harga menyebabkan kurva permintaan uang bergeser ke kiri; pergeseran MD ini menyebabkan r turun sehingga Y turun.
- Bank Sentral mengurangi jumlah uang beredar, yang menyebabkan kurva penawaran uang bergeser ke kiri; pergeseran MS ini menyebabkan r naik sehingga Y turun.

Suatu negara mengalami resesi akibat permintaan agregat (AD) yang tidak memadai. Pemerintah percaya bahwa menggeser kurva AD ke kanan sebesar 3.600 triliun akan mengakhiri resesi. Jika bank sentral menyesuaikan jumlah uang beredar untuk menjaga tingkat suku bunga tetap, investasi tetap tidak berubah, dan *marginal propensity to consume* (MPC) adalah $\frac{2}{3}$, berapa banyak penurunan pajak pemerintah yang diperlukan untuk mengakhiri resesi?

- a. 1.200 triliun.
- b. 1.800 triliun.
- c. 2.400 triliun.
- d. 3.600 triliun.

Jika bank sentral dalam pertanyaan sebelumnya justru mempertahankan jumlah uang beredar dan membiarkan tingkat suku bunga berubah, maka perubahan permintaan agregat yang dihasilkan dari penurunan pendapatan pajak pemerintah tersebut akan menjadi

- a. lebih besar
- b. Sama
- c. lebih kecil tapi masih positif
- d. Negatif

Misalkan penurunan suku bunga menyebabkan pengangguran berkurang dan tingkat output meningkat di negara Peninvilla. Kebijakan yang dapat diambil oleh otoritas moneter Peninvilla untuk mengatasi persoalan *overheating* yang mungkin terjadi dalam ekonomi adalah

- a. mengurangi pembelian obligasi pemerintah
- b. mengurangi pengeluaran untuk proyek infrastruktur public
- c. menurunkan rasio cadangan wajib bank
- d. menaikkan tingkat pajak

Esai:

1. Pada kuartal pertama 2025 Amerika Serikat (AS) telah mengalami perlambatan ekonomi. Pemerintah AS kemudian memberikan stimulus fiskal dalam bentuk subsidi sektoral ke sektor pertanian dan infrastruktur.
 - a. Misalkan, diketahui bahwa PDB riil AS adalah USD24 triliun, *marginal propensity to consume* (MPC) adalah 40%, dan stimulus fiskal adalah sebesar USD1,8 triliun. Berapakah PDB riil AS setelah terjadi *multiplier effect* dari stimulus fiskal?
 - b. Jika otoritas moneter di AS (the Fed) berupaya mempertahankan tingkat bunga dengan menaikkan uang beredar untuk mengimbangi kenaikan permintaan uang, apakah *crowding-out effect* dari stimulus fiskal pemerintah akan terjadi? Jelaskan jawaban Anda menggunakan diagram permintaan agregat (AD) dan keseimbangan di pasar uang.

2. Misalkan Negeri Pariangan meningkatkan pengeluaran pemerintahnya sebesar Rp200 miliar serta menaikkan pajak sebesar Rp200 miliar. Diketahui MPC di Negeri Pariangan adalah 0,80 dan tidak ada efek *crowding out* atau akselerator.
 - a. Hitung dan jelaskan efek dari gabungan kedua kebijakan tersebut terhadap permintaan agregat
 - b. Gambarkan dampak kebijakan gabungan tersebut secara grafis terhadap ekonomi Pariangan
3. John Maynard Keynes berpikir bahwa perilaku perekonomian dalam jangka pendek dipengaruhi oleh apa yang ia sebut sebagai “animal spirits.”

Yang ia maksud adalah bahwa para pelaku bisnis kadang merasa optimis terhadap perekonomian sehingga melakukan banyak investasi, dan di waktu lain merasa pesimis sehingga mengurangi pengeluaran investasinya. Jelaskan bagaimana fluktuasi investasi tersebut dapat menyebabkan fluktuasi dalam PDB riil dan tingkat harga.
4. Jelaskan mengapa tingkat suku bunga merupakan *opportunity cost* dari memegang uang tunai. Apa manfaat dari memegang uang tunai